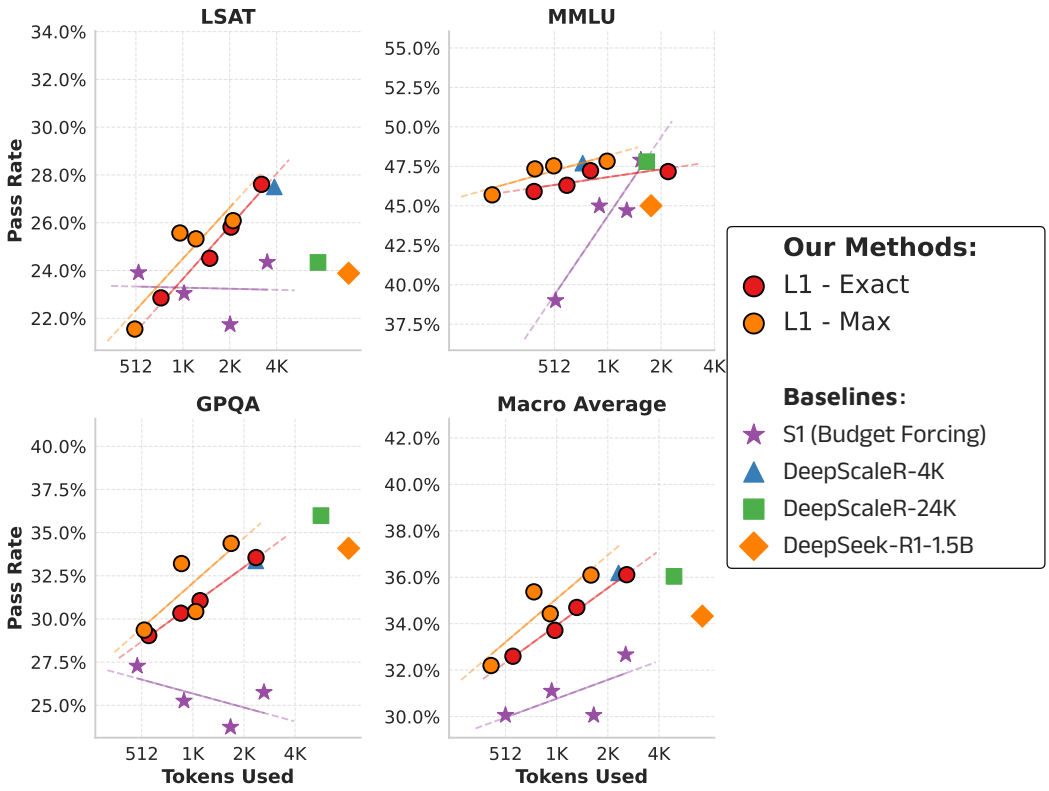

Figure 3: **Out-of-domain (OOD) generalization of L1.** L1 generalizes its length control to out-of-domain tasks, showing linear scaling of pass rate (y-axis) over generation length (x-axis) matching or exceeding performance of various baselines.

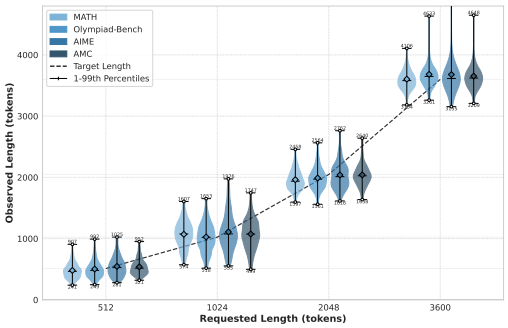

Figure 4: **Length control precision** across different datasets. The plot show distribution of response lengths at different requested lengths. Our model maintains consistent length control, with output lengths closely matching the requested lengths.

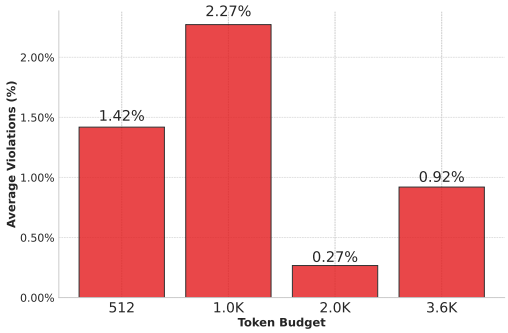

Figure 5: **Budget Violation Rates** across different datasets. L1-Max maintains less than 2.5% budget soft violation rate on across different token budgets, with lower violations at higher token lengths. On average the violation rate of mere 1.3%.

L1 follows length constraints with high precision. We quantitatively assess L1's ability to follow length constraints across various mathematical reasoning datasets. As shown in Figure 4, our model maintains consistent control across all token budgets (512, 1024, 2048, and 3600 tokens), with observed output lengths usually closely matching the requested lengths. Further, in Figure 11, we show the mean error: $(\frac{E_{x\sim D}[n_{generated}]-n_{gold}}{n_{gold}})$, which captures the

Model	AIME		MATH		AMC		Olympid-Bench		Average	
	# Tokens	Accuracy	# Tokens	Accuracy	# Tokens	Accuracy	# Tokens	Accuracy	# Tokens	Accuracy
L1-Max	401	5.0	299	70.8	431	47.1	410	33.4	385	39.1
L1-Short	422	12.1	321	75.4	395	46.7	390	36.0	382	42.6
Qwen-1.5	877	7.7	554	73.4	791	47.5	784	35.4	752	41.0
L1-Exact	880	9.2	583	70.6	801	49.7	770	**35.5**	758	41.2
L1-Max	858	**11.9**	541	**78.3**	757	**55.9**	722	**38.5**	720	**46.2**
Llama-3.3-70B	1024	3.8	573	74.9	838	48.6	859	37.3	824	41.2
L1-Exact	983	10.6	583	70.6	801	49.7	913	36.9	820	41.9
L1-Max	1059	**15.0**	631	**79.4**	921	**59.2**	853	**40.5**	866	**48.5**
GPT-4o	867	10.0	591	78.4	990	51.8	911	**42.4**	840	45.6
L1-Exact	880	10.0	583	70.6	1060	51.3	913	36.9	859	42.2
L1-Max	858	**11.9**	631	**79.4**	921	**59.2**	853	40.5	816	**47.8**

Table 1: **Short CoT performance across math benchmarks.** L1-Max surpasses its non-reasoning base model and competes with frontier models under identical short CoT lengths.

average deviation from target lengths across the dataset. The figure demonstrates that mean error is low: close to 3% for all math reasoning datasets. Although OOD datasets exhibit predictably higher errors (20-40%), these remain preferable over uncontrolled prompting. Further Analysis in Appendix A.2 demonstrates that larger errors primarily appear at higher token budgets on tasks like MMLU, where the longer chain of thoughts is mostly unnecessary. Additionally, in Appendix A.1, we show that error can be further reduced significantly with extended RL training.

Further, for L1-Max, we show the budget violation rates in Figure 5. In particular, we compute soft violation rates: $|n_{generated} - n_{gold}| > 500$. The violation rates are low, ranging from 0.3% to 2.3% across different token budgets. The results demonstrate that L1-Max can be reliably used for ensuring length control.

Long CoT Models are secretly Strong Short CoT Models. Given L1's strong performance at lower token budgets, we conducted a focused evaluation comparing it to both its base non-reasoning model (Qwen-2.5-1.5B-Instruct) and significantly larger non-reasoning models (GPT-4o and Llama-3.3-70B) at comparable *generation lengths*. Table 1 presents these results, showing that L1 consistently outperforms or matches all models across all datasets despite using equivalent token budgets. Further, on average, L1 is 5% better than its non-reasoning counterpart, and even outperforms GPT-4o by 2% on average. We refer to these models as Short Reasoning Models (SRMs), given their shorter generation lengths, yet exhibiting similar patterns as full reasoning models (See Section 5).

This finding is remarkable, as to the best of our knowledge, this is the first demonstration that a 1.5B model can outperform frontier models such as GPT-4o, despite using the ***same generation length***. Overall, the results signify that with suitable RL training, long CoT models can be adaptively used as short CoT models, while significantly outperforming their base counterparts at the same generation length.

L1 employs distinct reasoning strategies at different token budgets. To understand how L1 changes its reasoning approach across different length constraints, we analyzed how frequently certain reasoning-related terms appear in outputs of different lengths. Specifically, we calculated the normalized occurrence rate of most common reasoning terms in 512-token outputs compared to 4096-token outputs, showing how the model's reasoning strategies shift when given different length constraints. Figure 6 organizes these keywords into 4 distinct reasoning patterns: "Self-Correction and Verification," "Exploration and Alternatives," "Context Setting," and "Conclusion Drawing."

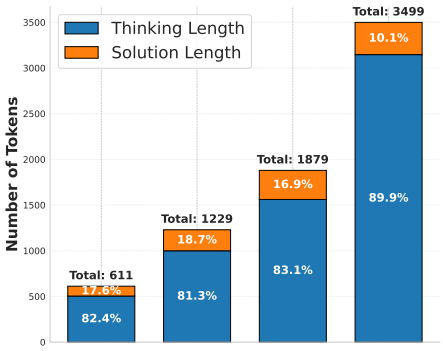

Figure 7: **Distribution of tokens for thinking vs. solution across chain-of-thought lengths.**

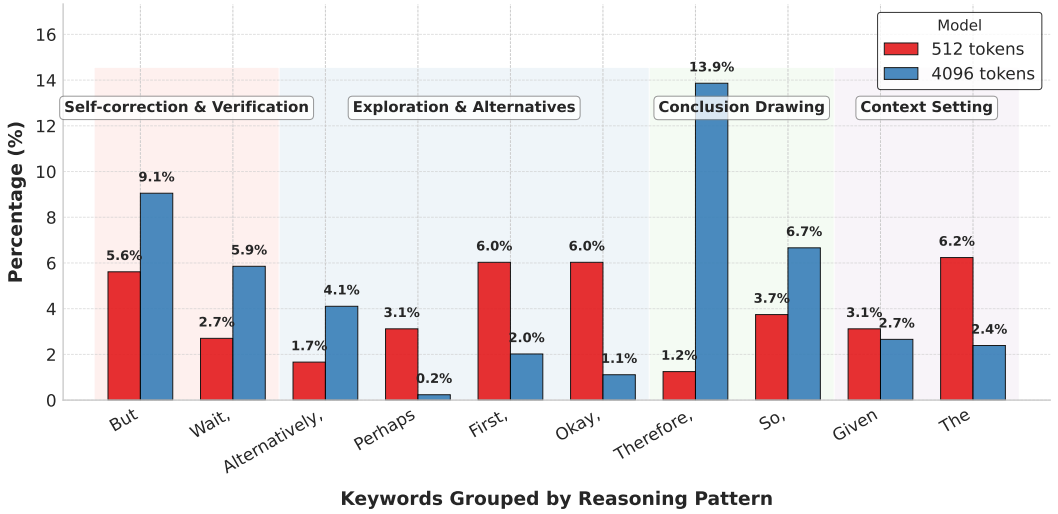

Figure 6: **Keyword usage by reasoning pattern at 512 vs. 4096 tokens.** L1 adaptively adjusts reasoning patterns based on requested token length.

Figure 6 shows that self-correction and verification keywords appear approximately twice as frequently in 4096-token outputs compared to 512-token outputs. Similarly, conclusion-drawing terms increase 2-10x with increased token budget, indicating more thorough solution validation and completion. Interestingly, most exploration-related keywords decrease in relative frequency at higher token counts, with "Alternatively" being a notable exception. Overall, we observe that smaller CoTs have reasoning patterns similar to their longer counterparts, but with changed relative frequencies that favor more self-verification and conclusion drawing in longer CoTs.

Further, Figure 7 shows the ratio of thinking tokens (those within `<think>` tags) to solution tokens for different generation lengths. We observe that the ratio is relatively stable across different generation lengths. This implies for shorter CoTs, the model generally provides short solutions (often just outputting the final answer), which helps save tokens. As generation length increases, we notice a stabilized response length in the last two bars, implying the model scales its thinking tokens without making the final solution overly verbose.

Additional Experiments and Analysis We further conduct several additional experiments and analysis to validate the robustness and generalizability of our approach across different configurations and model scales.

Reward Function Ablations. We systematically evaluate alternative reward function formulations for L1-Max in Appendix A.7 to understand the impact of different objective designs on both performance and length adherence. Our ablation study examines three variants: a single-objective approach using only the maximum length constraint, an additive formulation combining correctness and length penalties, and a sigmoid-based penalty function. While all variants achieve similar budget violation rates, the additive formulation exhibits undesirable behavior by collapsing to extremely short chain-of-thought sequences that trivially satisfy constraints but severely degrade reasoning quality. Our chosen multiplicative formulation in Equation 2 strikes an optimal balance, maintaining strong performance while ensuring robust adherence to length constraints.

Sampling Parameter Robustness. Our analysis in Appendix A.4 demonstrates that L1's length control capabilities remain stable across varying decoding parameters. When evaluated across temperature settings from 0.0 to 1.0, the model maintains consistent performance with mean length deviations of only 5.6-6.3% from target constraints. This robustness is particularly important for practical deployment, as it ensures reliable length control regardless of the specific sampling strategy employed. The observed average token lengths (1765-1782) closely match the theoretical expectation of 1796 tokens when averaged across our standard